PYTHON MCQ PRACTICE

1. What is the output of `print(type([]))`?
   A. `<class 'list'>`
   B. `<type 'list'>`
   C. `<list>`
   D. `list`
   ✅ Answer: A

2. Which keyword is used for function in Python?
   A. `function`
   B. `fun`
   C. `def`
   D. `define`
   ✅ Answer: C

3. Which of the following is not a core data type in Python?
   A. Tuples
   B. Lists
   C. Class
   D. Dictionary
   ✅ Answer: C

4. What is the result of `2 ** 3 ** 2`?
   A. 64
   B. 512
   C. 256
   D. 36
   ✅ Answer: B (Because it is `2 ** (3 ** 2)` = `2 ** 9` = 512)

5. Which of the following is correct about Python?
   A. Python is a low-level language
   B. Python is only used for web development
   C. Python supports object-oriented programming
   D. Python doesn't support functions
   ✅ Answer: C

6. How do you insert comments in Python code?
   A. `/* comment */`

B. `<!-- comment -->`
C. `# comment`
D. `// comment`
✅ Answer: C

7. Which of these is the slicing operator in Python?
   A. `::`
   B. `[]`
   C. `:`
   D. `()`
   ✅ Answer: C

8. What does `len("Python")` return?
   A. 5
   B. 6
   C. 7
   D. 0
   ✅ Answer: B

9. Which of the following is immutable?
   A. List
   B. Dictionary
   C. Set
   D. Tuple
   ✅ Answer: D

10. What is the output of `'a' + 'bc'`?
    A. abc
    B. a bc
    C. a+bc
    D. a
    ✅ Answer: A

11. Which function is used to find the length of a list in Python?
    A. `count()`
    B. `length()`
    C. `size()`
    D. `len()`

✅ Answer: D

12. Which of these is not a loop structure in Python?
    A. `for`
    B. `foreach`
    C. `while`
    D. `nested for`
    ✅ Answer: B

13. What is the output of `bool(0)`?
    A. True
    B. False
    ✅ Answer: B

14. What will `range(3)` return?
    A. [1, 2, 3]
    B. [0, 1, 2]
    C. [0, 1, 2, 3]
    D. Error
    ✅ Answer: B

15. Which statement is used to stop a loop?
    A. `exit`
    B. `break`
    C. `stop`
    D. `return`
    ✅ Answer: B

16. What does the `pass` keyword do?
    A. Skips the function
    B. Exits the program
    C. Does nothing
    D. Raises an error
    ✅ Answer: C

17. Which of the following is a correct function definition?
    A. `def myFunc:`
    B. `function myFunc()`

C. `def myFunc():`
D. `define myFunc():`
✅ Answer: C

18. Which keyword is used to return a value from a function?
A. `yield`
B. `break`
C. `return`
D. `pass`
✅ Answer: C

19. Which of the following data types is not iterable?
A. list
B. string
C. int
D. tuple
✅ Answer: C

20. What is the output of: `print(5 == 5.0)`?
A. False
B. True
✅ Answer: B

21. How do you add an element to a list?
A. `list.add()`
B. `list.append()`
C. `list.insert()`
D. `list.extend()`
✅ Answer: B

22. Which method removes the last item in a list?
A. `remove()`
B. `delete()`
C. `pop()`
D. `discard()`
✅ Answer: C

23. Which data type does not allow duplicate values?
    A. list
    B. tuple
    C. dict
    D. set
    ✅ Answer: D

24. What is the output of `set([1,2,2,3])`?
    A. [1,2,3]
    B. {1,2,3}
    C. (1,2,3)
    D. Error
    ✅ Answer: B

25. What is a dictionary in Python?
    A. Ordered collection of values
    B. Collection of key-value pairs
    C. Immutable set
    D. List of tuples
    ✅ Answer: B

26. Which method returns keys from a dictionary?
    A. `dict.keys()`
    B. `dict.allKeys()`
    C. `dict.getKeys()`
    D. `dict.items()`
    ✅ Answer: A

27. Which operator is used to check membership?
    A. `==`
    B. `!=`
    C. `in`
    D. `is`
    ✅ Answer: C

28. What is the result of `{'a':1, 'b':2}['a']`?
    A. a
    B. 1
    C. 2

D. Error
✅ Answer: B

29. How to update a value in a dictionary?
    A. `dict.update()`
    B. `dict.edit()`
    C. `dict['key'] = new_value`
    D. `dict.modify()`
    ✅ Answer: C

30. Which of the following will convert a list to a tuple?
    A. `list()`
    B. `tuple()`
    C. `set()`
    D. `dict()`
    ✅ Answer: B

31. Which keyword is used to create a class in Python?
    A. `class`
    B. `def`
    C. `struct`
    D. `object`
    ✅ Answer: A

32. How do you create an object in Python?
    A. `obj = new ClassName()`
    B. `obj = ClassName()`
    C. `obj = create ClassName()`
    D. `obj = Class.new()`
    ✅ Answer: B

33. What is `self` in Python classes?
    A. Reference to the class
    B. Global variable
    C. Instance of the object
    D. Keyword for return
    ✅ Answer: C

34. Which method is called when an object is created?
    A. `__init__`
    B. `__create__`
    C. `constructor`
    D. `__new__`
    ✅ Answer: A

35. What is inheritance?
    A. A way to define variables
    B. A way to execute loops
    C. A mechanism to derive a class from another
    D. A way to import modules
    ✅ Answer: C

36. What is method overriding?
    A. Defining two methods with the same name
    B. Changing a method's return type
    C. Rewriting a parent method in a child class
    D. None of the above
    ✅ Answer: C

37. What is encapsulation?
    A. Wrapping data and methods into one unit
    B. Inheriting multiple classes
    C. Hiding data outside the class
    D. Both A and C
    ✅ Answer: D

38. What is polymorphism?
    A. Changing variable types
    B. One function, many forms
    C. One class, many objects
    D. One object, many classes
    ✅ Answer: B

39. What is multiple inheritance?
    A. Inheriting one class from multiple classes
    B. Inheriting one class from one class
    C. Repeating the base class
    D. Inheriting a class multiple times

✅ Answer: A

40. Which function is used to check the class of an object?
    A. `class()`
    B. `typeof()`
    C. `isinstance()`
    D. `objclass()`
    ✅ Answer: C

41. What is a lambda function?
    A. Named function
    B. Anonymous function
    C. Recursive function
    D. Class method
    ✅ Answer: B

42. What is the output of `lambda x: x+10` when x=5?
    A. 10
    B. 5
    C. 15
    D. Error
    ✅ Answer: C

43. Which of these is used for exception handling?
    A. `try-except`
    B. `catch`
    C. `error`
    D. `raise`
    ✅ Answer: A

44. Which keyword is used to raise an exception?
    A. `throw`
    B. `raise`
    C. `except`
    D. `error`
    ✅ Answer: B

45. What is the output of `type(lambda x: x)`?
    A. `<class 'function'>`
    B. `<function>`
    C. `<lambda>`
    D. `<class 'lambda'>`
    ✅ Answer: A

46. What does `*args` do?
    A. Passes multiple keyword arguments
    B. Passes a list
    C. Passes multiple positional arguments
    D. Creates a tuple
    ✅ Answer: C

47. What does `**kwargs` do?
    A. Collects keyword arguments into a dictionary
    B. Unpacks a list
    C. Collects all arguments
    D. Collects positional arguments
    ✅ Answer: A

48. Which module is used to handle dates in Python?
    A. `datetime`
    B. `calendar`
    C. `date`
    D. `time`
    ✅ Answer: A

49. What does `map()` do?
    A. Loops through a string
    B. Applies a function to each item in an iterable
    C. Filters a list
    D. Returns indexes
    ✅ Answer: B

50. Which of these libraries is used for data analysis?
    A. `NumPy`
    B. `Pandas`

C. `Matplotlib`
D. `Flask`
✅ Answer: B

51. What is the output of `print(type([]))`?
A) `<class 'list'>` ✅
B) `<type 'list'>`
C) `list`
D) `[]`

52. Which of the following is a mutable data type?
A) str
B) tuple
C) list ✅
D) int

53. What is the result of `3 * 1 ** 3`?
A) 27
B) 9
C) 3 ✅
D) 1

54. Which keyword is used to define a function in Python?
A) function
B) def ✅
C) func
D) define

55. What is the output of `print(bool(0))`?
A) True
B) False ✅
C) 0
D) Error

56. Which of the following is used to define a block of code in Python?
A) Curly braces
B) Parentheses
C) Indentation ✅
D) Quotes

57. **What is the output of `len("Hello")`?**
    A) 4
    B) 5 ✅
    C) 6
    D) Error

58. How do you insert COMMENTS in Python?
    A) // Comment
    B) /* Comment */
    C) # Comment ✅
    D) -- Comment

59. Which operator is used for exponentiation?
    A) ^
    B) ** ✅
    C) *
    D) //

60. **What is the result of `10 // 3`?**
    A) 3 ✅
    B) 3.33
    C) 4
    D) Error

61. Which data type is immutable?
    A) List
    B) Dictionary
    C) Set
    D) Tuple ✅

62. Which method can be used to convert a string to lowercase?
    A) lower() ✅
    B) down()
    C) toLower()
    D) lowerCase()

63. Which of the following is not a valid variable name?
    A) my_var
    B) _myvar
    C) my-var ❌

D) myvar1

64. How do you get the data type of a variable?
A) typeof()
B) gettype()
C) type() ✅
D) dtype()

65. **What is the result of `bool("False")`?**
A) False
B) True ✅
C) None
D) Error

66. **What will `print("apple" > "Apple")` return?**
A) True ✅
B) False
C) Error
D) None

67. What is the default return value of a function that doesn't explicitly return anything?
A) None ✅
B) 0
C) False
D) Undefined

68. **How do you create a list in Python?**
A) `list = ()`
B) `list = []` ✅
C) `list = {}`
D) `list = <>`

69. Which of the following is not a keyword in Python?
A) pass
B) eval ✅
C) assert
D) nonlocal

70. Which function can be used to get user input?

A) scanf()
B) cin>>
C) input() ✅
D) read()

71. **What does `len([1, 2, [3, 4]])` return?**

A) 2
B) 3 ✅
C) 4
D) Error

72. How do you start a multiline string in Python?

A) ''' or """ ✅
B) {{}}
C) []
D) //

73. **What is the output of `print(2 in [1, 2, 3])`?**

A) True ✅
B) False
C) 2
D) Error

74. Which keyword is used for loop in Python?

A) loop
B) for ✅
C) iterate
D) whiledo

75. **What is the output of `print(0.1 + 0.2 == 0.3)`?**

A) True
B) False ✅
C) 0.3
D) Error

76. What will `range(3)` return?

A) [0, 1, 2] ✅
B) [1, 2, 3]
C) [0, 1, 2, 3]

D) None

26. What is the output of the code: `for i in range(0, 10, 2): print(i)`?
    A) 0 2 4 6 8 ✅
    B) 0 1 2 3 4
    C) 2 4 6 8 10
    D) 0 2 4 6 8 10

27. What is the purpose of `continue` in a loop?
    A) Exit loop
    B) Skip current iteration ✅
    C) Restart loop
    D) Pause loop

28. Which of these can define a function with default parameters?
    A) `def foo(a, b=1):` ✅
    B) `def foo(a=1, b):`
    C) `def foo(a=1 b=2):`
    D) `function foo(a, b=1):`

29. What is the output of `print("abc".replace("a", "z"))`?
    A) abc
    B) zbc ✅
    C) abz
    D) Error

31. What is the output of `print("Hello"[-1])`?
    A) H
    B) o ✅
    C) l
    D) Error

32. Which of the following is not a loop in Python?
    A) for
    B) while
    C) do-while ✅
    D) None

33. How many times will this loop run? `for i in range(3):`
    A) 2

B) 3 ✅
C) 4
D) Infinite

34. How do you call a function named `my_func`?
A) call my_func
B) my_func[]
C) my_func() ✅
D) def my_func

35. What is the output of `sum([1, 2, 3])`?
A) 6 ✅
B) [1, 2, 3]
C) Error
D) 123

36. What is the result of `",".join(["a", "b", "c"])`?
A) a,b,c ✅
B) abc
C) a,c,b
D) ['a','b','c']

37. Which of these creates a function?
A) `create function f()`
B) `def f():` ✅
C) `function f():`
D) `f() define:`

38. What does `str(123)` return?
A) 123
B) '123' ✅
C) Error
D) None

39. What does `int("10") + float("5.5")` evaluate to?
A) 15.5 ✅
B) 105.5
C) Error

D) 10.0

40. What does `print("ab" * 3)` output?
    A) ab
    B) abab
    C) ababab ✅
    D) Error

41. Which function returns the ASCII of a character?
    A) char()
    B) ascii()
    C) ord() ✅
    D) asc()

42. What does `range(1,5,2)` return?
    A) [1,2,3,4,5]
    B) [1,2,3,4]
    C) [1,3] ✅
    D) [1,2]

43. What is the purpose of `pass`?
    A) Terminates loop
    B) Continues loop
    C) Does nothing ✅
    D) Returns from function

44. What is printed by `print(type(lambda x: x))`?
    A) function
    B) lambda
    C) `<class 'function'>` ✅
    D) None

45. What is the output of `print([1,2,3][::-1])`?
    A) [1,2,3]
    B) [3,2,1] ✅
    C) [2,1,3]
    D) Error

46. How to reverse a string s in Python?
    A) `s.reverse()`
    B) `reversed(s)`
    C) `s[::-1]` ✅
    D) `reverse(s)`

47. What is the output of `print(2 ** 3 ** 2)`?
    A) 512 ✅
    B) 64
    C) 256
    D) 36

48. What does `print([1,2]+[3,4])` output?
    A) [1,2,3,4] ✅
    B) [1,2][3,4]
    C) Error
    D) [1,2+3,4]

49. What will be the output of `print(min([4,1,9]))`?
    A) 1 ✅
    B) 4
    C) 9
    D) Error

50. Which of the following functions is used to read a file?
    A) open() ✅
    B) readfile()
    C) inputfile()
    D) file.open()

51. What does `open('file.txt', 'w')` do?
    A) Opens for reading
    B) Opens for appending
    C) Opens for writing (truncates if exists) ✅
    D) Opens in binary

52. Which keyword is used for exception handling?
    A) catch
    B) handle

C) except ✅
D) error

53. Which exception is raised for division by zero?
A) ZeroDivision ✅
B) ArithmeticError
C) DivideError
D) MathError

54. What is the output of `set([1,2,2,3])`?
A) [1,2,2,3]
B) {1,2,3} ✅
C) {1,2,2,3}
D) [1,2,3]

55. What will `d = {1: 'a', 2: 'b'}; print(d[1])` return?
A) 'a' ✅
B) 'b'
C) 1
D) Error

56. How do you handle multiple exceptions?
A) Multiple try blocks
B) Multiple except blocks ✅
C) Multiple raise blocks
D) Multiple catch blocks

57. What is the output of `bool([])`?
A) True
B) False ✅
C) None
D) Error

58. What data type does `input()` return in Python 3?
A) int
B) str ✅
C) bool
D) float

59. Which of the following is not a core Python data type?
    A) Lists
    B) Dictionary
    C) Tuples
    D) Class ✅

60. What does `list("abc")` return?
    A) ['abc']
    B) ['a','b','c'] ✅
    C) ['a b c']
    D) ['c','b','a']

61. Which method is a constructor in Python?
    A) create
    B) start
    C) init ✅
    D) construct

62. What does `is` compare in Python?
    A) Values
    B) Identity ✅
    C) Type
    D) Both value and type

63. What is `__str__` used for?
    A) String operations
    B) Memory representation
    C) String representation ✅
    D) System call

64. Which module provides regular expressions?
    A) string
    B) regex
    C) re ✅
    D) pattern

65. What is the output of `type([]) == list`?
    A) True ✅
    B) False
    C) Error

D) None

66. Which keyword is used to inherit a class?
    A) inherit
    B) parent
    C) class Child(Parent): ✅
    D) subclass

67. Which scope does a variable declared in a function belong to?
    A) Local ✅
    B) Global
    C) Built-in
    D) Outer

68. How do you import only `sqrt` from `math` module?
    A) `import math.sqrt`
    B) `from math import sqrt` ✅
    C) `import sqrt from math`
    D) `math.import(sqrt)`

69. Which operator is used to check object type?
    A) ==
    B) is
    C) isinstance ✅
    D) typecheck

70. What is the output of `None == 0`?
    A) True
    B) False ✅
    C) None
    D) Error

71. What is the output of `print("a" in "apple")`?
    A) True ✅
    B) False
    C) None
    D) Error

91. How to install external Python libraries?
    A) `pip install package_name` ✅
    B) `install package_name`
    C) `python -m install`
    D) `setup install`

93. Which module in Python is used for JSON operations?
A) xml
B) csv
C) json ✅
D) pickle

94. What is the output of `print(type(range(5)))`?
A) `<class 'list'>`
B) `<class 'range'>` ✅
C) `<class 'tuple'>`
D) `<class 'generator'>`

95. What will `list(map(str, [1, 2, 3]))` return?
A) ['1', '2', '3'] ✅
B) ['1 2 3']
C) [1, 2, 3]
D) Error

96. How do you create an anonymous function in Python?
A) def lambda():
B) lambda ✅
C) anon()
D) def anon():

97. Which of these is NOT a Python built-in function?
A) zip()
B) filter()
C) reduce() ✅
D) map()

98. How do you comment multiple lines in Python?
A) /* comment */
B) # on each line ✅

C) <!-- comment -->
D) ''' comment ''' only for docstrings

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99. What does `list(zip([1, 2], ['a', 'b']))` return?
A) [(1, 'a'), (2, 'b')] ✅
B) [(1, 2), ('a', 'b')]
C) [(1, 'b'), (2, 'a')]
D) Error

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100. Which of the following is used to create a virtual environment in Python?
A) venv ✅
B) virtualenv
C) env
D) pyenv

THEORY QUESTIONS

1. What is Python?

Answer: Python is a high-level, interpreted, general-purpose programming language known for its readability and dynamic typing.

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2. What are Python's key features?

Answer: Easy syntax, interpreted, dynamically typed, automatic memory management, large standard library, supports multiple paradigms (procedural, OOP, functional).

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3. What are Python's data types?

Answer: int, float, bool, str, list, tuple, dict, set, frozenset, complex, NoneType.

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4. What is the difference between a list and a tuple?

Answer: Lists are mutable (can be changed), tuples are immutable (cannot be changed).

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5. How do you declare a variable in Python?

Answer: Just assign a value: `x = 5`

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6. What is PEP 8?

Answer: PEP 8 is Python's style guide for writing clean, readable code.

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7. How do you write comments in Python?

Answer: Use # for single-line comments and triple quotes `'''` or `"""` for multiline docstrings.

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8. What is indentation in Python?

Answer: Python uses indentation (spaces or tabs) to define code blocks instead of braces `{}`.

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9. What are functions in Python?

Answer: Functions are reusable blocks of code defined with the `def` keyword.

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10. What is a lambda function?

Answer: An anonymous one-line function defined with `lambda`, e.g., `lambda x: x + 1`.

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11. **What are *args and kwargs in functions?

Answer: `*args` allows passing a variable number of positional arguments; `**kwargs` allows passing variable keyword arguments.

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12. What is list comprehension?

Answer: A concise way to create lists, e.g., `[x*2 for x in range(5)]`.

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13. How to handle exceptions in Python?

Answer: Use `try-except` blocks.

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14. What is the difference between `is` and ==?

Answer: `is` checks identity (same object), == checks equality of values.

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15. What are Python modules and packages?

Answer: Modules are `.py` files with Python code; packages are directories with modules and a special `__init__.py`.

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16. How to import modules?

Answer: Using `import module_name` or `from module_name import function`.

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17. What is a Python dictionary?

Answer: A key-value data structure, unordered and mutable.

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18. How to iterate over a dictionary?

Answer: Using `.items()`, `.keys()`, or `.values()` methods.

---

19. What is a Python set?

Answer: An unordered collection of unique elements.

---

20. What is the difference between deep copy and shallow copy?

Answer: Shallow copy copies object references, deep copy copies objects recursively.

---

21. How do you create a virtual environment?

Answer: Using `python -m venv env_name`.

---

22. What is a generator in Python?

Answer: A function that yields values one at a time, using `yield`.

---

23. What is the difference between list, tuple, and set?

Answer: Lists are ordered and mutable, tuples ordered and immutable, sets unordered and unique elements.

---

24. How to open and read a file?

Answer: Use `open('filename', 'r')` and `.read()` or `.readlines()`.

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25. What are Python decorators?

Answer: Functions that modify the behavior of other functions.

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26. Explain Python's memory management.

Answer: Python uses reference counting and garbage collection for memory management.

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27. What is the difference between `pass`, `continue`, and `break`?

Answer: `pass` does nothing, `continue` skips to the next loop iteration, `break` exits the loop.

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28. What is a Python iterator?

Answer: An object with `__iter__()` and `__next__()` methods to iterate over data.

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29. How does Python handle memory leaks?

Answer: Python uses garbage collection to free unused memory but leaks can happen with circular references without proper handling.

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30. What are Python's built-in data structures?

Answer: List, tuple, dict, set.

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31. What is slicing?

Answer: Extracting parts of sequences like lists or strings using `[start:stop:step]`.

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32. How do you reverse a list?

Answer: Using `list[::-1]` or `list.reverse()`.

---

33. How to concatenate two lists?

Answer: Using + operator or `.extend()` method.

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34. What are Python comprehensions?

Answer: Concise ways to create lists, sets, dicts: list comprehensions, set comprehensions, dict comprehensions.

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35. What is the GIL?

Answer: Global Interpreter Lock; it allows only one thread to execute Python bytecode at a time.

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36. How to handle multiple exceptions?

Answer: Use `except (Exception1, Exception2):`.

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37. What is the difference between `filter()` and `map()`?

Answer: `filter()` selects elements, `map()` transforms elements.

---

38. What is a Python class?

Answer: A blueprint for creating objects with attributes and methods.

---

39. What is inheritance?

Answer: A class can inherit methods and properties from another class.

---

40. What are instance variables and class variables?

Answer: Instance variables belong to objects; class variables belong to the class itself.

---

41. What is method overriding?

Answer: Subclass provides its own implementation of a method inherited from the parent class.

---

42. What are magic methods?

Answer: Special methods like `__init__`, `__str__`, `__repr__`, `__add__` that start and end with double underscores.

---

43. What is duck typing?

Answer: Python's dynamic typing means if it walks like a duck and quacks like a duck, it's a duck.

---

44. What are Python iterables?

Answer: Objects that can return an iterator, e.g., lists, tuples, strings.

---

45. How do you install external packages?

Answer: Using `pip install package_name`.

---

46. What is a Python namespace?

Answer: A mapping between names and objects.

---

47. What are global and local variables?

Answer: Global variables are defined outside functions, local variables inside functions.

---

48. How to declare a global variable inside a function?

Answer: Use the `global` keyword.

---

49. What is the purpose of `self` in class methods?

Answer: Refers to the instance calling the method.

---

50. What are Python closures?

Answer: Functions that remember values from enclosing scopes.

---

51. What is the difference between `del` and `remove()`?

Answer: `del` removes by index or variable, `remove()` removes by value.

---

52. What is the use of `with` statement?

Answer: To simplify exception handling and ensure resource cleanup, like closing files.

---

53. What is the difference between `append()` and `extend()`?

Answer: `append()` adds a single item, `extend()` adds elements from another iterable.

---

54. How do you convert a string to int?

Answer: Using `int('123')`.

---

55. What is the difference between Python 2 and 3?

Answer: Python 3 has better Unicode support, `print` is a function, and many other improvements.

---

56. What is a Python bytecode?

Answer: Intermediate representation of Python code, executed by the Python virtual machine.

---

57. What are Python's built-in functions for sequence types?

Answer: `len()`, `max()`, `min()`, `sum()`, `sorted()`.

---

58. What is a Python set?

Answer: An unordered collection of unique elements.

---

59. How to check if a key exists in a dictionary?

Answer: Using `key in dict`.

---

60. What is the difference between `pop()` and `remove()`?

Answer: `pop()` removes by index and returns the element; `remove()` removes by value.

---

61. What is multithreading in Python?

Answer: Running multiple threads (lightweight processes) within a single process.

---

62. What are Python iterators?

Answer: Objects with a `__next__()` method that produce successive values.

---

63. What are Python generators?

Answer: Special iterators created with functions using `yield`.

---

64. What is the difference between `range()` and `xrange()`?

Answer: In Python 2, `xrange()` is a generator version of `range()`. Python 3 has only `range()` which behaves like `xrange()`.

---

65. What are metaclasses?

Answer: Classes of classes that define how classes behave.

---

66. What is the difference between shallow copy and deep copy?

Answer: Shallow copy copies references, deep copy copies objects recursively.

---

67. What is pickling and unpickling?

Answer: Serializing and deserializing Python objects to/from byte streams.

---

68. What are Python's truthy and falsy values?

Answer: Values considered true/false in conditionals, e.g., `0`, `None`, `''` are falsy.

---

69. What is the purpose of the `zip()` function?

Answer: Combines multiple iterables into tuples.

---

70. What is monkey patching?

Answer: Dynamically modifying a class or module at runtime.

---

71. What are Python's logical operators?

Answer: `and`, `or`, `not`.

---

72. What is the difference between mutable and immutable objects?

Answer: Mutable objects can be changed, immutable cannot.

---

73. How do you generate random numbers?

Answer: Using the `random` module.

---

74. What is the use of the `enumerate()` function?

Answer: Adds a counter to an iterable.

---

75. What are Python's built-in types for sequences?

Answer: List, tuple, range.

---

76. How to copy a list?

Answer: Using `list.copy()` or `list[:]`.

---

77. What are Python's comparison operators?

Answer: `==`, `!=`, `>`, `<`, `>=`, `<=`.

---

78. How do you check the Python version?

Answer: `python --version` or `import sys; print(sys.version)`.

---

79. What is the difference between `input()` and `raw_input()`?

Answer: In Python 2, `raw_input()` reads strings, `input()` evaluates input; Python 3 only has `input()`.

---

80. What is the purpose of the `assert` statement?

Answer: Used for debugging, checks a condition and raises AssertionError if false.

---

81. How to reverse a string?

Answer: Using slicing: `string[::-1]`.

---

82. What is the difference between `filter()` and list comprehension?

Answer: `filter()` applies a function to filter elements, list comprehension can do filtering and transformation.

---

83. What is the difference between == and `is`?

Answer: == checks value equality, `is` checks object identity.

---

84. How can you manage memory in Python?

Answer: Python has automatic memory management using reference counting and garbage collection.

---

85. What is the use of the `super()` function?

Answer: To call a method from the parent class.

---

86. What are Python's numeric types?

Answer: int, float, complex.

---

87. What is the use of the `dir()` function?

Answer: Lists the attributes and methods of an object.

---

88. What are Python's truthy values?

Answer: Non-zero numbers, non-empty collections, `True`.

---

89. How do you swap two variables in Python?

Answer: `a, b = b, a`

---

90. What is the difference between `break` and `return`?

Answer: `break` exits a loop, `return` exits a function.

---

91. What is the purpose of the `global` keyword?

Answer: To modify a global variable inside a function.

---

92. How can you generate a random integer between 1 and 10?

Answer: `random.randint(1, 10)`

---

93. What does the `pass` statement do?

Answer: Does nothing, used as a placeholder.

---

94. How do you convert a string to a list?

Answer: Using `list(string)`.

---

95. What is the use of the `any()` function?

Answer: Returns True if any element of the iterable is true.

---

96. How do you check if a string starts with a substring?

Answer: Using `str.startswith()`.

---

97. What is the output of `bool([])`?

Answer: False.

---

98. What is the difference between `remove()` and `del`?

Answer: `remove()` deletes by value, `del` deletes by index or variable.

---

99. What is a metaclass?

Answer: A class of a class that defines how a class behaves.

---

100. How to handle multiple exceptions in one block?

Answer: Using `except (Exception1, Exception2):`.